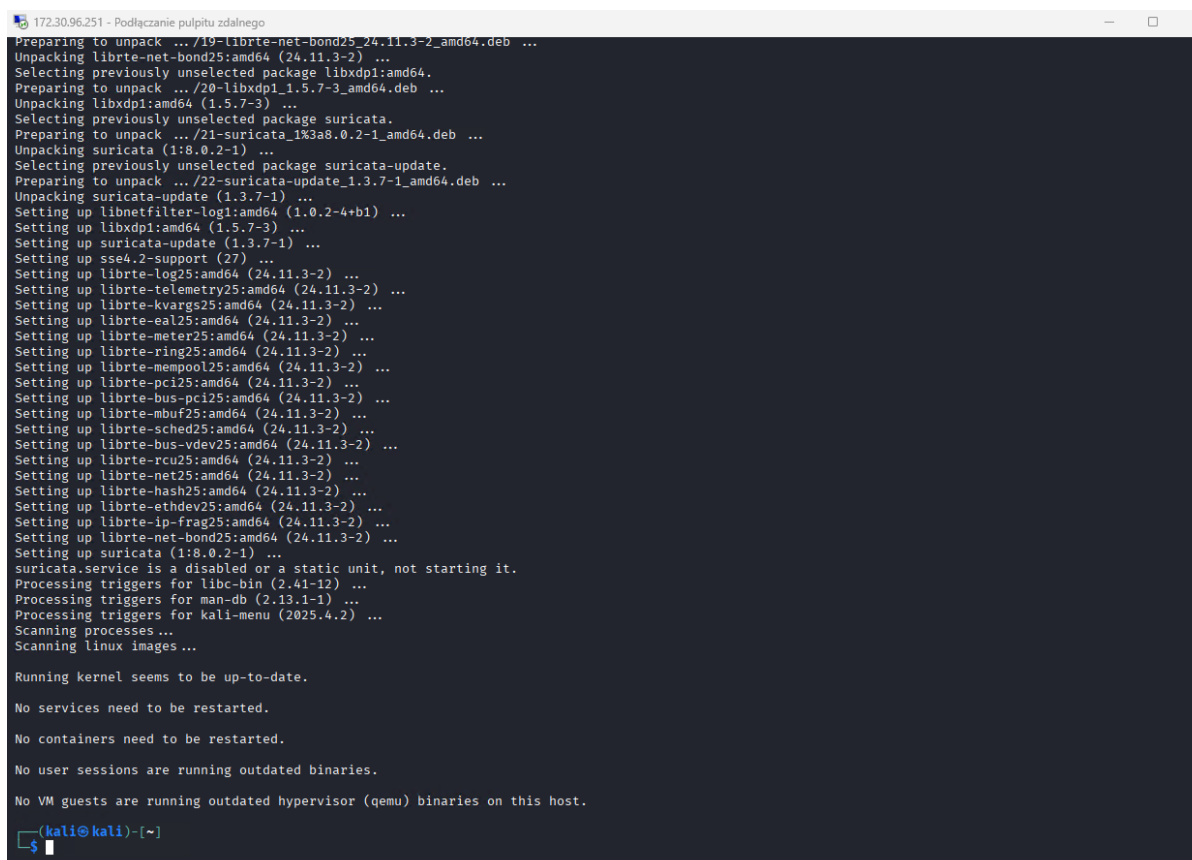




```
172.30.96.251 - Podłączenie pulpitu zdalnego
Preparing to unpack .../19-librtre-net-bond25_24.11.3-2_amd64.deb ...
Unpacking librtre-net-bond25:amd64 (24.11.3-2) ...
Selecting previously unselected package libxdp1:amd64.
Preparing to unpack .../20-libxdp1_1.5.7-3_amd64.deb ...
Unpacking libxdp1:amd64 (1.5.7-3) ...
Selecting previously unselected package suricata.
Preparing to unpack .../21-suricata_1%3a8.0.2-1_amd64.deb ...
Unpacking suricata (1:8.0.2-1) ...
Selecting previously unselected package suricata-update.
Preparing to unpack .../22-suricata-update_1.3.7-1_amd64.deb ...
Unpacking suricata-update (1.3.7-1) ...
Setting up libnetfilter-log1:amd64 (1.0.2-4+b1) ...
Setting up libxdp1:amd64 (1.5.7-3) ...
Setting up suricata-update (1.3.7-1) ...
Setting up sse4.2-support (27) ...
Setting up librtre-log25:amd64 (24.11.3-2) ...
Setting up librtre-telemetry25:amd64 (24.11.3-2) ...
Setting up librtre-kvargs25:amd64 (24.11.3-2) ...
Setting up librtre-eal25:amd64 (24.11.3-2) ...
Setting up librtre-meter25:amd64 (24.11.3-2) ...
Setting up librtre-ring25:amd64 (24.11.3-2) ...
Setting up librtre-mempool25:amd64 (24.11.3-2) ...
Setting up librtre-pci25:amd64 (24.11.3-2) ...
Setting up librtre-bus-pci25:amd64 (24.11.3-2) ...
Setting up librtre-mbuf25:amd64 (24.11.3-2) ...
Setting up librtre-sched25:amd64 (24.11.3-2) ...
Setting up librtre-bus-vdev25:amd64 (24.11.3-2) ...
Setting up librtre-rcu25:amd64 (24.11.3-2) ...
Setting up librtre-net25:amd64 (24.11.3-2) ...
Setting up librtre-hash25:amd64 (24.11.3-2) ...
Setting up librtre-ethdev25:amd64 (24.11.3-2) ...
Setting up librtre-ip-frag25:amd64 (24.11.3-2) ...
Setting up librtre-net-bond25:amd64 (24.11.3-2) ...
Setting up suricata (1:8.0.2-1) ...
suricata.service is a disabled or a static unit, not starting it.
Processing triggers for libc-bin (2.41-12) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.2) ...
Scanning processes ...
Scanning linux images ...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

(kali@kali)-[~]
$
```

- Action: Installation of the Suricata package and management tools (suricata-update) on Kali Linux using the apt package manager.
- Technical Objective: Preparation of the Intrusion Detection and Prevention System (IDS/IPS) engine for deployment within the network infrastructure.
- Screenshot Analysis: The terminal output documents the extraction and installation process of core components, including suricata, suricata-update, and supporting libraries (e.g., librtre for DPDK/network subsystems). The successful execution of system triggers is visible.
- Success Criterion: The packages were successfully installed on the system. Executing the suricata -V command following this step confirms the engine's readiness for operational mode configuration (passive/inline).

```
172.30.96.251 - Podłączanie pulpitu zdalnego
ERROR: suricata: unit tests not enabled. Make sure to pass --enable-unittests to configure when building.

(kali@kali)-[~]
$ sudo suricata-update
1/12/2025 -- 11:05:00 -- <Info> -- Using data-directory /var/lib/suricata.
1/12/2025 -- 11:05:00 -- <Info> -- Using Suricata configuration /etc/suricata/suricata.yaml
1/12/2025 -- 11:05:00 -- <Info> -- Using /etc/suricata/rules for Suricata provided rules.
1/12/2025 -- 11:05:00 -- <Info> -- Found Suricata version 8.0.2 at /usr/bin/suricata.
1/12/2025 -- 11:05:00 -- <Info> -- Loading /etc/suricata/suricata.yaml
1/12/2025 -- 11:05:00 -- <Info> -- Disabling rules for protocol pgsq
1/12/2025 -- 11:05:00 -- <Info> -- Disabling rules for protocol modbus
1/12/2025 -- 11:05:00 -- <Info> -- Disabling rules for protocol dnp3
1/12/2025 -- 11:05:00 -- <Info> -- Disabling rules for protocol enip
1/12/2025 -- 11:05:00 -- <Info> -- No sources configured, will use Emerging Threats Open
1/12/2025 -- 11:05:00 -- <Info> -- Fetching https://rules.emergingthreats.net/open/suricata-8.0.2/emerging.rules.tar.gz.
100% - 5181078/5181078
1/12/2025 -- 11:05:08 -- <Info> -- Done.
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/app-layer-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/decoder-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/dhcp-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/dnp3-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/dns-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/files.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/http2-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/http-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/ipsec-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/kerberos-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/modbus-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/mqtt-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/nfs-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/ntp-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/quic-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/rfb-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/smb-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/smtp-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/ssh-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/stream-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Loading distribution rule file /etc/suricata/rules/tls-events.rules
1/12/2025 -- 11:05:08 -- <Info> -- Ignoring file 61af9651a149b484162bdbdb08823d48/rules/emerging-deleted.rules
1/12/2025 -- 11:05:11 -- <Info> -- Loaded 62369 rules.
1/12/2025 -- 11:05:11 -- <Info> -- Disabled 13 rules.
1/12/2025 -- 11:05:11 -- <Info> -- Enabled 0 rules.
1/12/2025 -- 11:05:11 -- <Info> -- Modified 0 rules.
1/12/2025 -- 11:05:11 -- <Info> -- Dropped 0 rules.
1/12/2025 -- 11:05:12 -- <Info> -- Enabled 136 rules for flowbit dependencies.
1/12/2025 -- 11:05:12 -- <Info> -- Backing up current rules.
1/12/2025 -- 11:05:12 -- <Info> -- Writing rules to /var/lib/suricata/rules/suricata.rules: total: 62369; enabled: 46553; added: 62369; removed 0; modified 0
1/12/2025 -- 11:05:12 -- <Info> -- Writing /var/lib/suricata/rules/classification.config
1/12/2025 -- 11:05:12 -- <Info> -- Testing with suricata -T.
1/12/2025 -- 11:05:51 -- <Info> -- Done.

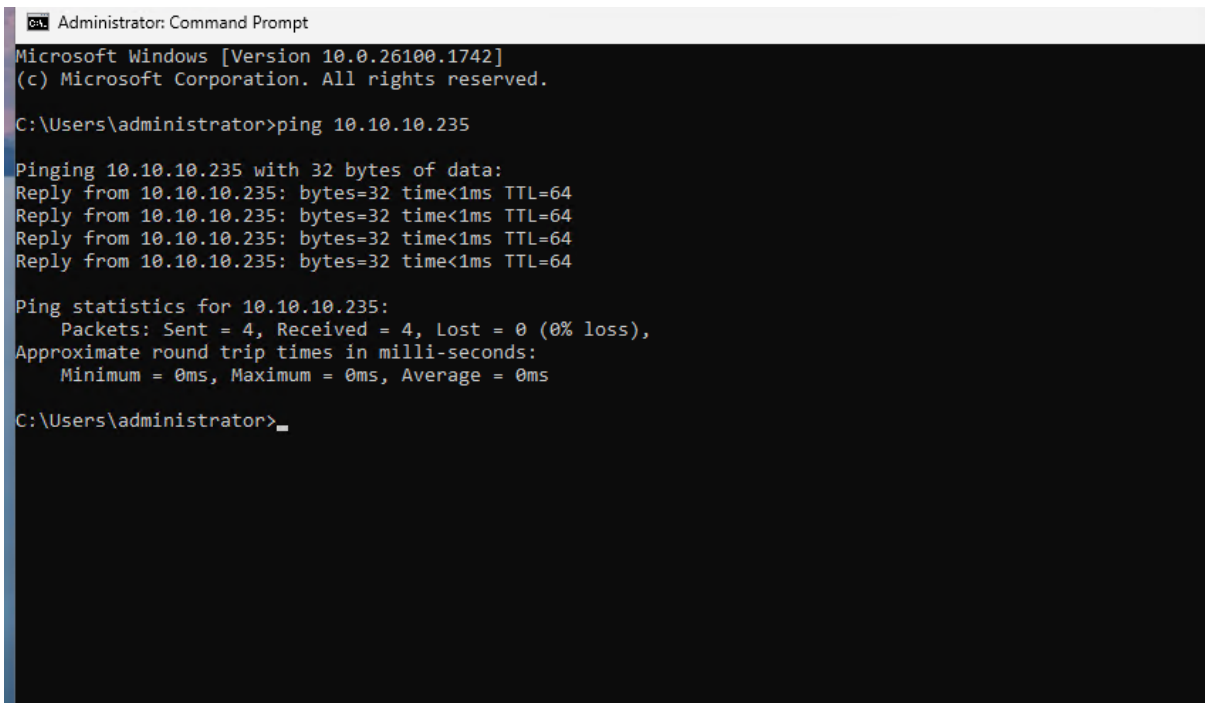
(kali@kali)-[~]
$
```

- Action: Execution of the suricata-update tool to download and verify current rulesets.
- Technical Objective: Ensuring up-to-date coverage against known threats by retrieving signatures from the Emerging Threats Open ruleset.
- Screenshot Analysis: The system successfully connected to the Emerging Threats servers and downloaded the emerging.rules.tar.gz archive. The logs indicate that 62,369 rules were loaded, of which 46,553 were activated. A crucial element is the final configuration validation (Testing with suricata -T), which completed successfully.
- Success Criterion: Rule files were updated and verified for consistency, ensuring the system's readiness to detect current adversary TTPs (Tactics, Techniques, and Procedures).

```
- fast:
  enabled: yes
  filename: fast.log
  append: yes
  #filetype: regular # 'regular', 'unix_stream' or 'unix_dgram'

# Extensible Event Format (nicknamed EVE) event log in JSON format
- eve-log:
  enabled: yes
  filetype: regular #regular/syslog/unix_dgram/unix_stream/redis
  filename: eve.json
```

- Action: Editing the suricata.yaml configuration file within the log output definitions section.
- Technical Objective: Enabling event logging in formats essential for rapid triage and subsequent integration with SIEM solutions.
- Screenshot Analysis: Two key log types were enabled:
 1. fast.log: Providing a quick, human-readable overview of alerts.
 2. eve.json (EVE): An extended JSON format containing comprehensive metadata regarding network flows, DNS transactions, HTTP requests, and alerts, which is necessary for advanced event correlation.
- Success Criterion: The system correctly generates artifacts in the EVE JSON format, establishing the foundation for modern network observability.



```

Administrator: Command Prompt
Microsoft Windows [Version 10.0.26100.1742]
(c) Microsoft Corporation. All rights reserved.

C:\Users\administrator>ping 10.10.10.235

Pinging 10.10.10.235 with 32 bytes of data:
Reply from 10.10.10.235: bytes=32 time<1ms TTL=64
Reply from 10.10.10.235: bytes=32 time<1ms TTL=64
Reply from 10.10.10.235: bytes=32 time<1ms TTL=64
Reply from 10.10.10.235: bytes=32 time<1ms TTL=64

Ping statistics for 10.10.10.235:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\administrator>

```

- Action: Execution of a network connectivity test using the ping command from a Windows host to the target host.
- Technical Objective: Establishing a connectivity baseline prior to the activation of IPS blocking rules and verifying routing correctness within the lab environment.
- Screenshot Analysis: The Windows console displays four successful replies from the IP address 10.10.10.235 with zero packet loss.
- Success Criterion: Basic network connectivity is confirmed; the systems are ready to generate synthetic traffic designed to trigger IDS rule matches.

- Screenshot Analysis:
 - Top section (fast.log): Four entries for the ICMP protocol, triggered by a local rule (SID: 1000001), are visible, indicating traffic between hosts 10.10.10.100 and 10.10.10.235.
 - Bottom section (eve.json): The same traffic was recorded in a structured JSON format, where the "event_type": "alert" field and "signature": "ALERT" confirm consistency with the IDS mode.
 - The use of the grep command allows for the rapid identification of the attack source and destination (src_ip and dest_ip) without the need to manually search through the entire log files.
- Success Criterion: The visualization of alerts in both formats serves as definitive proof of a correct IDS configuration. It is confirmed that every policy violation attempt (in this case, ICMP traffic) generates a persistent artifact in the logs, which is crucial for accountability and incident analysis.